HIGH PRECISION ELECTROSTATIC ACTUATOR WITH NOVEL ELECTRODE DESIGN

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ABSTRACT

A novel electrode design of electrostatic surface drives, which can reduce the drives' force ripple considerably, is described in this paper. Actuators with large travel and highly precise motion control are required for many micro mechanical systems. Electrostatic surface drives are promising, since their structures are very simple and they easily possess large travel. However, surface drives often have large force ripple that prevents precise motion control. To suppress the force ripple, we propose to utilize skew technique for their electrodes. The analytical results show that utilizing the new electrode design can reduce force ripple by 75%, without a remarkable lowering of thrust. Reducing force ripple leads to smooth slider motions, higher driving efficiency, more quietness, and higher controllability which is experimentally confirmed using a newly fabricated prototype drive.

INTRODUCTION

Many micro mechanical systems require micro linear positioning devices with large travel and highly precise motion control. For such positioning devices, electrostatic linear surface drives are promising because of the following reasons: (1) their structures are simple enough to fabricate; (2) they can easily possess large travel; (3) they can possess high power density that leads to rapid responses.

Linear surface drives consist of two surfaces, a stator and a slider. In most cases, large attraction is generated between these two surfaces. In microsized actuators, supporting the slider against large attraction is difficult. Therefore, surface drives should be designed to minimize attraction. Applying different voltages in each surface at the same time is one of the most efficient designs to reduce attraction. Fig. 1 shows one example of such effective actuator designs that can increase thrust and decrease attraction. Besides, these designs allow a synchronous operation which makes controlling the drives easier. However, the drives with such designs possess largely rippled force generated at the same regular intervals as the electrode pitch. The force ripple makes difficult precise motion control such as positioning control with a resolution smaller than the electrode pitch.

On the other hand, the authors have developed high power electrostatic surface drives using flexible printed circuitry (FPC) films[2,3,4]. Their thrusts are more than several newtons with their weights of several grams. Power per weight ratios of those high power electrostatic drives exceed those of electromagnetic linear drives. Therefore, they are expected to be employed in various mechatronics systems, such as computer peripherals, robotic systems, and machine tools, instead of electromagnetic drives. However, these high power drives also possess large force ripples. The ripples must be eliminated for the mentioned applications.

In this paper, we propose skewing the electrodes of electrostatic surface drives for ripple reduction. For analyses and experiments, we employ the actuator design shown in Fig. 1 as a representative of linear surface drives. The analytical results show that skewing electrodes can cut 75% of the force ripple without a remarkable lowering of thrust. A newly fabricated prototype drive performs higher driving efficiency, more quietness, and higher controllability than ordinary ones that do not employ the skew technique.

DUAL EXCITATION MULTIPHASE ELECTROSTATIC DRIVE (DEMED)[3],[4]

Fig. 1 shows an electrostatic linear surface drive that is investigated in this paper. This drive is called *dual excitation multiphase electrostatic drive* (DEMED). DEMED consists of two flat parts, a slider and a stator. Each of them contains three-phase parallel electrodes. It is operated by applying a three-phase ac voltage to three poles of both stator and slider. The connections to stator and slider

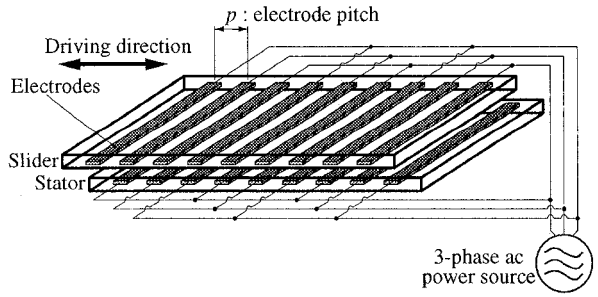


Fig. 1 Schematic view of *Dual Excitation Multiphase Electrostatic Drive* (DEMED), one of electrostatic linear surface drives.

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are in reversed order. Then potential distributions traveling in the opposite directions are excited on both surfaces. Electrostatic interactions between these two potential distributions actuate the slider. The velocity of slider is equal to the difference of the two distributions' velocities. This is expressed as:

$$v_s = 6 \cdot p \cdot f_{ac} \tag{1}$$

where v_s, p, and f_{ac} are velocity of slider, electrode pitch, and a frequency of an applied ac, respectively.

The thrust of DEMED is given as a function of slider position and phase of applied ac. Fig. 2 shows the thrust function of DEMED, calculated using *surface charge method* [3,4]. To plot Fig. 2, we focused on one line of the three-phase ac. The phase of that line is expressed as excitation phase. The slider position is the relative displacement of two poles to which that line is connected. According to Fig. 2, the thrust of DEMED greatly ripples even though sinusoidal signals are applied.

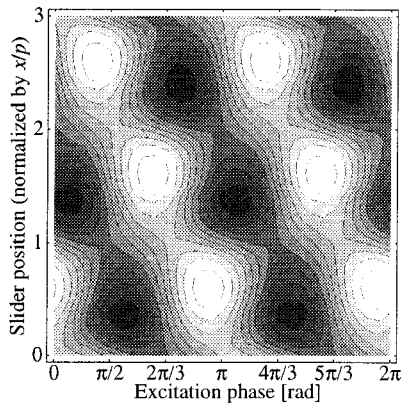


Fig. 2 Contour plot of thrust of DEMED. Light area expresses positive thrust; and dark area expresses negative thrust.

PROPOSAL AND ANALYSES OF ELECTRODE SKEW[5]

To reduce the ripple observed in Fig. 2, we consider skewing the drives' electrodes. The basic type of skew, which is firstly analyzed, is illustrated in Fig. 3. In this electrodes' arrangement, both electrodes of stator and slider are rotated in the same angle in reversed direction to each other, as to maintain the symmetry between stator and slider. We define l and θ_s as the width of electrode area and an angle between stator's and slider's electrodes, respectively. In the following analyses, l is constant. Therefore, electrode length is varied with θ_s as to keep l constant.

The effect of skew cannot be described with θ_s only. Therefore, we define skew factor F_s as:

$$F_s = \frac{2l}{p} \sin\frac{\theta_s}{2} \tag{2}$$

Skew factor is described by the number of the stator's electrodes intersected by a slider's electrode.

We derive the characteristics of skewed DEMED by the following consideration: divided into narrow strips, each strip of skewed DEMED is considered to act as a non-skewed DEMED, as shown in Fig. 4. By supposing a limit where each strip is infinitesimal, the thrust function T of skewed DEMED is carried out as:

$$T(\varphi,x) = \int_{x-\frac{F_s}{2}}^{x+\frac{F_s}{2}} \frac{T_0(\varphi,\xi)}{F_s} d\xi \tag{3}$$

where x is a slider position, φ is an excitation phase, function $T_0(\varphi,x)$ is the thrust function of non-skewed DEMED shown in Fig. 2. In case a subscript is attached to T, the subscript indicates the skew factor for that function. As an example of calculated functions, $T_{1.5}(\varphi,x)$, the thrust function for $F_s = 1.5$, is plotted in Fig. 5.

To compare the thrust function for each skew factor, we define three indices: T_m, T_s, and ripple rate r. Index T_m indicates the global maximum value of the thrust function. Index T_s is the value of saddle points, as shown in Fig. 5. DEMED can generate T_s at any slider position, while it can generate T_m only at certain points.

If DEMED operates with constant synchronous velocity, the following equation is formed.

$$\varphi = -\frac{1}{2}\frac{2\pi}{3p}x + \delta \tag{4}$$

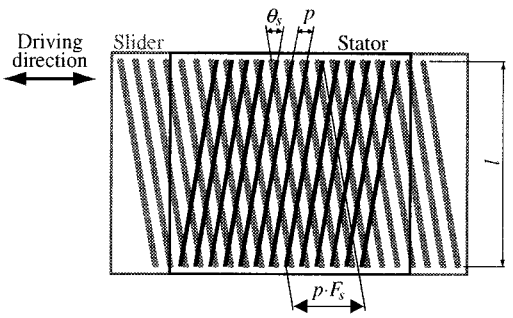


Fig. 3 Schematic top view of skewed DEMED. The electrodes of slider and stator are skewed in reversed direction to each other.

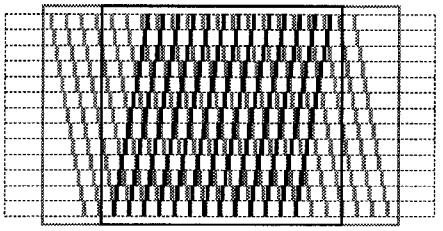


Fig. 4 Imaginary model for calculation. By dividing skewed electrodes into many narrow strips, each strip is supposed to act as one non-skewed DEMED.

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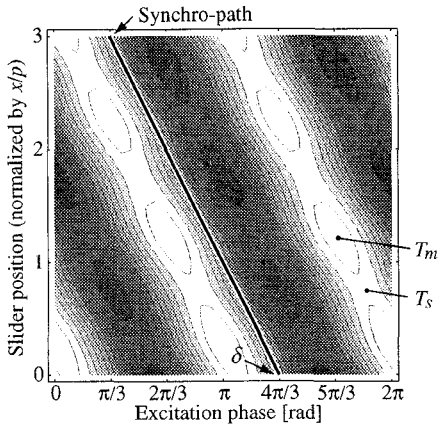


Fig. 5 Contour plot of $T_{1.5}(\varphi,x)$, thrust function of DEMED when skew factor $F_s = 1.5$.

This relationship can be expressed by a line in the thrust plot, as in Fig. 5. We call this line synchro-path. When DEMED generates the thrust of T_s, the synchro-path will pass the saddle points. Then ripple rate r is defined on that synchro-path as:

$$r = \frac{t_{max}(\delta_0) - t_{min}(\delta_0)}{t_{ave}(\delta_0)} \times 100 \qquad (5)$$

where $t_{min}(\delta)$ is the value nearest to 0, $t_{max}(\delta)$ is the maximum, and $t_{ave}(\delta)$ is the average of thrust, on the path defined by δ. δ_0 is determined as the path passes the saddle points.

Relationships between the indices T_m, T_s, and r and skew factor F_s are plotted in Fig. 6(a, b). Neglecting a drop of T_m, the best performance can be obtained near $F_s = 1.5$. At this point, T_s is maximized. The ripple rate is minimized and is a quarter of that of a non-skewed DEMED (skew factor of non-skewed DEMED is 0).

To keep the skew factor constant, the slider should not yaw. Since microsized actuators can hardly employ secure linear guides, the yawing moment of DEMED must possess certain characteristics to suppress the yawing motion of the slider.

Like the calculation of thrust, the yawing moment $M(\varphi,x)$ can be expressed as:

$$M(\varphi,x) = \int_{x-\frac{F_s}{2}}^{x+\frac{F_s}{2}} \frac{(\xi - x)\, T_0(\varphi,\xi)}{F_s} d\xi \qquad (6)$$

In equation (6), for increasing skew factor F_s, the acting moment $M(\varphi,x)$ is positive. To compare the moment function for each skew factor, we consider a synchro-path again. For each skew factor, we can calculate an average of the yawing moment $m_{ave}(\delta)$ and of the thrust $t_{ave}(\delta)$ on the synchro-path defined by a variable δ. By eliminating the variable δ from both functions, we obtain a function $m_{ave}(t_{ave})$ that indicates the relationship between the yawing moment and the thrust.

In Fig. 6(c), $m_{ave}(0)$ is plotted as a function of skew factor F_s. This is the average of the yawing moment when DEMED operates without load. It is found that DEMED can operate when F_s is 0, 4.3, or 7.4 only. Therefore a most effective skew factor, that is 1.5, cannot be utilized.

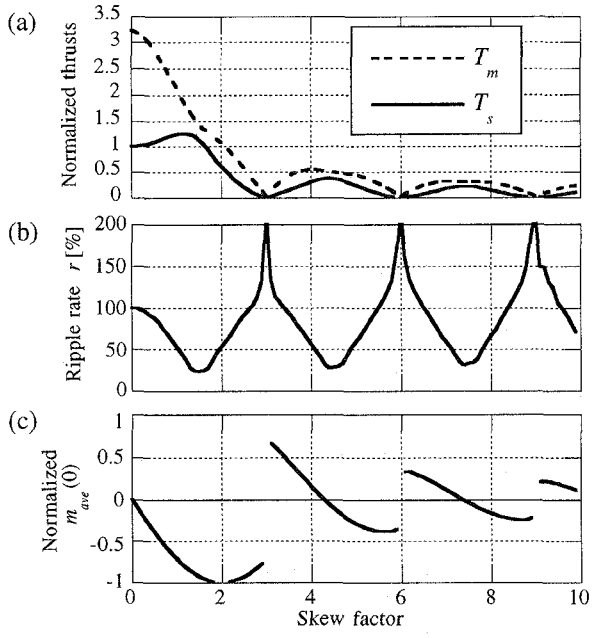


Fig. 6 Calculated performances of skewed DEMED

PROPOSAL AND ANALYSES OF WASHBOARD-SHAPED ELECTRODES

To utilize the skew factor of 1.5, we propose washboard-shaped electrodes shown in Fig. 7. A washboard-shaped electrode consists of several V-shaped electrodes. The number of connected V-shaped electrodes is expressed by variable v, as shown in Fig. 8. We call the drives with this electrodes' arrangement 'V-DEMED.' The thrust function of V-DEMED $T_v(\varphi,x)$ and $T_{1.5}(\varphi,x)$ are almost the same. Their magnitudes only are different, since operating directions are slightly different between V-DEMED and

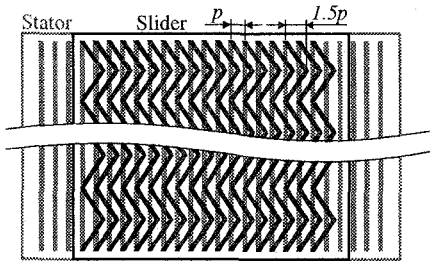


Fig. 7 Schematic top view of V-DEMED, utilizing washboard-shaped electrodes for slider.

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skewed DEMED. The difference is expressed by the angle θ_{diff}:

$$\frac{l}{2v}\sin 2\theta_{diff} = 1.5p \tag{7}$$

Then the thrust function of V-DEMED $T_V(\varphi,x)$ is

$$T_V(\varphi,x) = \cos\theta_{diff} \cdot T_{1.5}(\varphi,x) \tag{8}$$

However, the total electrode length l is usually far longer than electrode pitch p. Therefore, $\cos\theta_{diff} \approx 1$, unless v is quite large.

To suppress the yawing, washboard-shaped electrodes are symmetric to the axis of the driving direction. Therefore, the yawing moment is obviously zero when the slider does not yaw. However, when the slider yaws because of disturbances, a yawing moment may act on the slider. When the slider yaws, thrust $T_{yaw}(\varphi,x)$ and yawing moment $M_{yaw}(\varphi,x)$ are expressed as:

$$T_{yaw}(\varphi,x) = \int_{-l_1}^{l_2} \frac{T_0(\varphi,E(\xi))}{l_1+l_2} d\xi \tag{9}$$

$$M_{yaw}(\varphi,x) = \int_{-l_1}^{l_2} \frac{\xi\, T_0(\varphi,E(\xi))}{l_1+l_2} d\xi \tag{10}$$

where function $E(\xi)$ expresses the electrode shape of the yawed slider by assuming a stator electrode to be an axis,

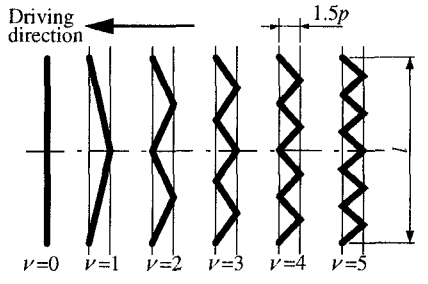


Fig. 8 Various types of washboard-shaped electrodes. Variable v expresses the number of connected V-shaped electrodes, except that $v = 0$ means parallel electrodes.

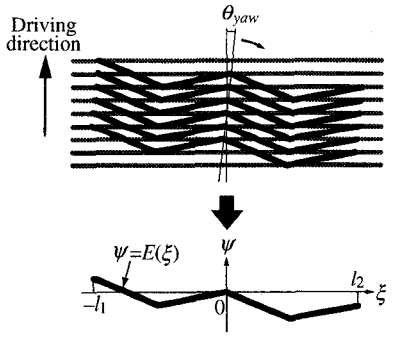


Fig. 9 A function $E(\xi)$ expresses a shape of a electrode of a yawed slider by assuming the stator's electrode to be an axis.

as shown in Fig. 9. The functions $m_{ave}(\delta)$ and $t_{min}(\delta)$ can be derived from $M_{yaw}(\varphi,x)$ and $T_{yaw}(\varphi,x)$. From those two functions, $m_{ave}(t_{min})$ can be derived by eliminating the variable δ. Then, yawing stiffness s_{yaw} are defined as:

$$s_{yaw}(t_{min}) = -\frac{m_{ave}(t_{min})}{\theta_{yaw}} \tag{11}$$

where θ_{yaw} are angle of yaw. The stiffness $s_{yaw}(t_{min})$ is plotted in Fig. 10. Since the electrode designs are not symmetric to the axis perpendicular to the driving direction, m_{ave} depends on the direction of thrust. Especially, smaller v results in larger dependency. If v is greater than 4, the dependency is negligibly small, and the stiffness is always positive.

FABRICATION AND EXPERIMENTS

We fabricated V-DEMED of $v = 4$. Fig. 11 shows their stator and slider. Their electrode pitch p is 200μm, and

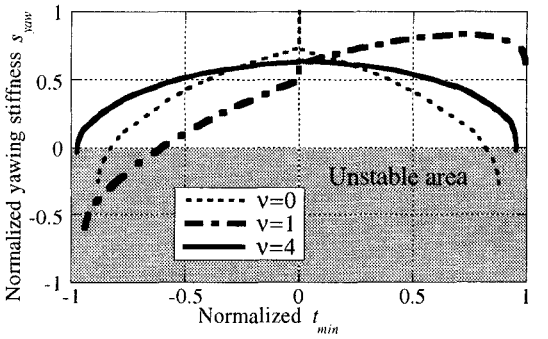


Fig. 10 Plots of $s_{yaw}(t_{min})$ for $v = 0$, 1, and 4, calculated under conditions of $\theta_{yaw} = 0.03°$ and $l = 480p$. If s_{yaw} is negative, the operation will not be stable. The thrust t_{min} is positive in the direction shown in Fig. 8 and 9.

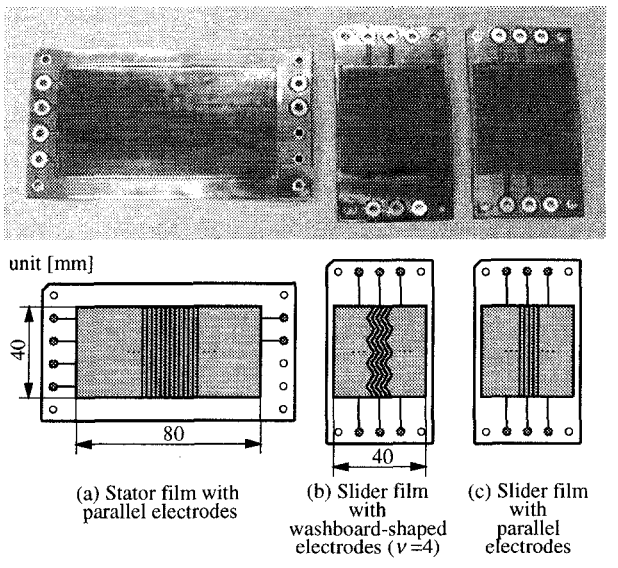


Fig. 11 Fabricated stator and slider FPC films. V-DEMED consists of stator (a) and slider (b), non-skewed DEMED consists of stator (a) and slider (c).

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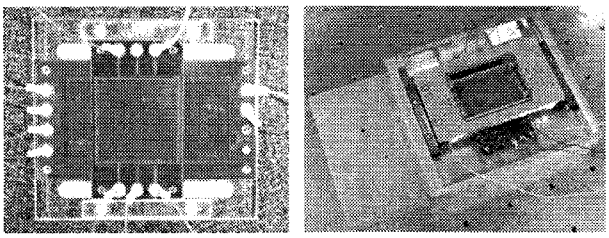
the width of electrode area l is 40mm. They are fabricated using FPC technique. Fig. 12(a) shows a basic setup of V-DEMED. In this setup, a stator and a slider are simply stacked, and the slider is not guided. Glass beads with a diameter of 20μm are inserted between the stator and the slider. The stator and the slider are immersed in dielectric liquid with a relative permittivity of 1.86. Electrodes of stator and slider face each other over an area of 1600mm^2. Normal DEMED was also fabricated sharing the same design in order to compare their performances.

Fig 13 shows the slider's motion of both DEMEDs operated in an open loop system. V-DEMED performs smooth motion, while non-skewed DEMED stepped at an interval of p.

To measure thrust force, we utilized another setup shown in Fig 12(b). In this setup, eight pairs of a stator and a slider are stacked and combined to enlarge the thrust. The combined sliders are guided using linear guides. The glass beads and the dielectric liquid are also utilized in this setup. The following experiments are carried out using this setup.

Fig. 14 shows the thrusts of both DEMEDs measured by simple weight pull-up experiments. The performance of each drive is almost the same, but especially at low speed, V-DEMED performs better. That is because T_s is dominant at low operation speed, while T_m is dominant at high operation speed.

Energy consumptions at 30mm/s are shown in Fig. 15. Before 0.4s, DEMEDs pull up a weight of 500g, and lower the weight after 0.4s. The efficiency of V-DEMED is 1.2 times of that of non-skewed DEMED when they are pulling up the weight. During lowering the weight, energy consumption of V-DEMED is less than half of that of non-skewed DEMED. We also measured energy consumptions at a speed of 15mm/s. At this speed, the efficiency of V-DEMED was 29% while that of non-skewed DEMED was 21%. The efficiency of V-DEMED was 1.4 times of that of non-skewed DEMED. This means, the improvement of efficiency is greater at low speed. In these experiments, the efficiency is not as high as compared to other actuators. That is because of less operation speed. In the past experiments, the efficiency of 70% was recorded for non-skewed DEMED at a speed of 154mm/s. For V-DEMED, the efficiency is expected to be as high at such a high speed.



(a) 1-layer setup (b) 8-layers setup

Fig. 12 Drive setups.

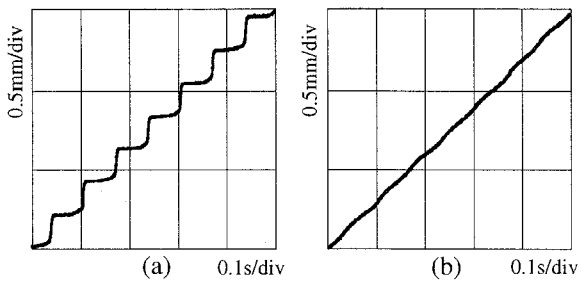
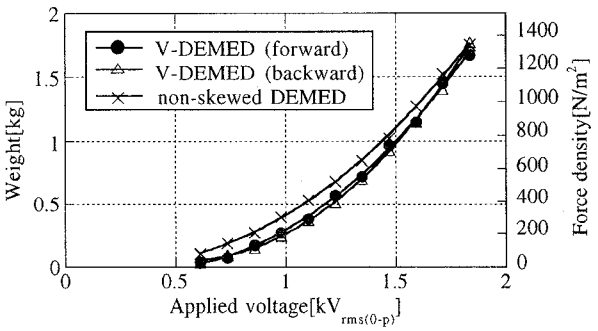
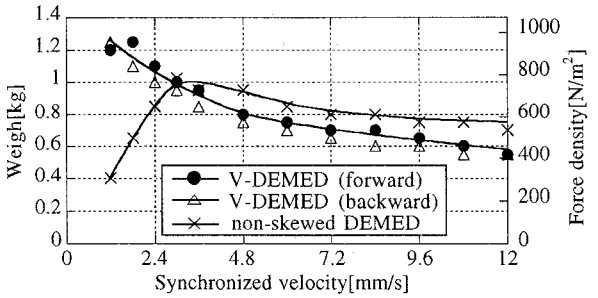


Fig. 13 Slider's motions of both DEMEDs in open loop operation, measured using 1-layer setup. Plot (a) is that of non-skewed DEMED, and (b) is that of V-DEMED. Applied voltages are 1.8kV$_{rms(o-p)}$ in both plots.



(a) Weight vs. applied voltage at 3mm/s



(b) Weight vs. operation speed at 1.5kV$_{rms(0-p)}$

Fig. 14 Plots of weight that both DEMEDs can pull up in open loop operation. These plots are meausured using 8-layers setup.

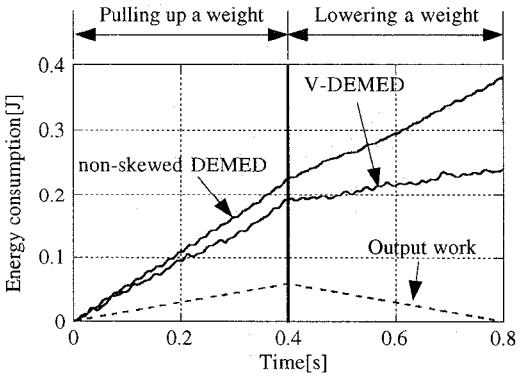


Fig. 15 Energy consumption of both DEMEDs in open loop operation when pulling up and lowering a weight of 500g. Speed is 30mm/s and voltage is 1.5kV$_{rms(0-p)}$.

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Vibrations of both DEMEDs are shown in Fig. 16. Applying skewing electrodes technique successfully reduced the vibration of DEMED. As a result, we also observed reduced noise, although this has not been measured quantitatively.

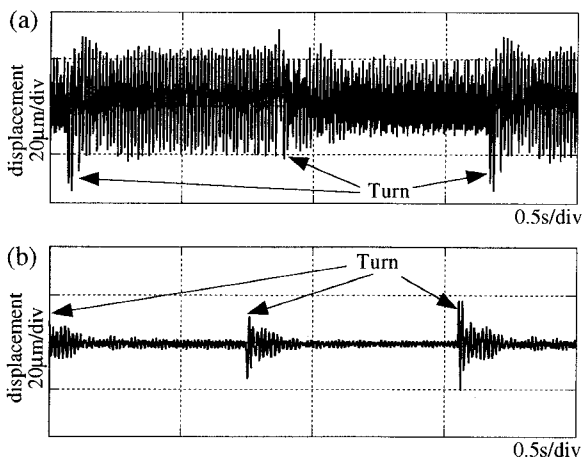


Fig. 16 Vibrations of both DEMEDs in open loop operation, measured at 30mm/s and 1.8kV$_{rms(0\text{-}p)}$ using 8-layers setup. Plot (a) is that of non-skewed DEMED, and (b) is of V-DEMED.

FEEDBACK CONTROL

To confirm the improved controllability of V-DEMED, feedback control is investigated using the 8-layers setup. PID controller built in DSP (TMS320C30) is utilized. The control cycle time of DSP is 40μs.

The thrust function of V-DEMED is simply given as the following equation, if we ignore the ripple.

$$T_v = \kappa_t V_0^2 \sin(2\delta) \tag{12}$$

where κ_t is a constant determined from the size of the drive. Then, applied voltages are determined by equation (4) and the following:

$$V_0 = \sqrt{\frac{t_{ref}}{\kappa_t}} \tag{13}$$

where t_{ref} is thrust reference given by the PID controller. To detect the slider position, a linear scale with a resolution of 1μm is mounted on the slider.

Fig. 17 shows results of feedback control. These diagrams show the path errors, when ramp position of 8.3mm/s is commanded to the controller. In case of V-DEMED, the path error is comparably small, while the path error of non-skewed DEMED is rippled at an interval of p.

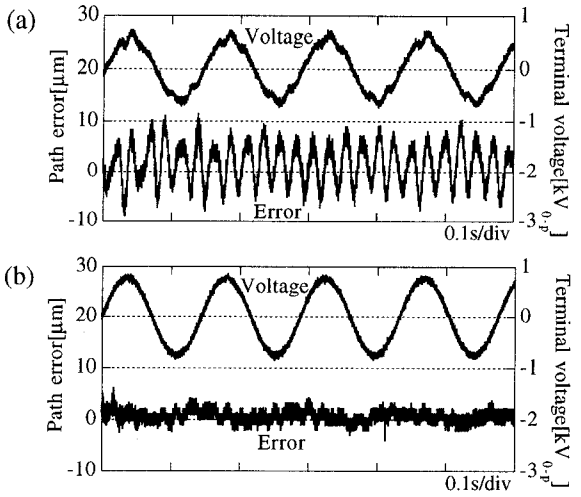


Fig. 17 Path errors and contolled voltages, driven at 8.3mm/s in a PID feedback system. Plot (a) is of non-skewed DEMED, and (b) is of V-DEMED.

CONCLUSION

To eliminate force ripple of electrostatic linear surface drives, we proposed to utilize skew on their electrodes. The effects of skew were analyzed. According to the analyses, the best performance could be obtained when electrodes were skewed by a skew factor F_s =1.5. However, it has also been shown that such skew is not stable because of slider's yawing. Then, we proposed washboard-shaped electrodes that can stabilize the yawing of the slider. A fabricated prototype has performed satisfactory smooth motion. Besides, efficiency and controllability were improved and vibration was reduced considerably, compared to the drive without this skew technique.

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